

The empirical recurrence for OEIS A235561: recovering a conjecture that is not written down

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Abstract

OEIS A235561 records “Empirical recurrence of order 67”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 90$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 67, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 67 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235561 is “Number of $(n+1) \times (2+1)$ 0..2 arrays with the difference of the upper and lower median value of each 2×2 subblock in lexicographically nondecreasing order rowwise and columnwise.”. It has offset 1 and begins

537, 8345, 113391, 1383617, 15633605, 165947729, 1680360507, 16368643767, ...

The entry states:

Empirical recurrence of order 67 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:39:01, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 227$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 90$ of the 227, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 90$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 454$ exact terms of the model returns order 67 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	28	c_{18}	67863530343164	c_{35}	−84778825344597931172	c_{52}	24022850244168571
c_2	−164	c_{19}	320815542368838	c_{36}	−58794404352237880616	c_{53}	−7353137855677372
c_3	−2330	c_{20}	−889623731586916	c_{37}	208728056285268582376	c_{54}	−8129591300360595
c_4	29621	c_{21}	−2099995408258286	c_{38}	48562438214149863232	c_{55}	4364454515576301
c_5	32810	c_{22}	8418883141785062	c_{39}	−406741387626536549520	c_{56}	1594965726329674
c_6	−1835349	c_{23}	9564875522735594	c_{40}	47277909765855203520	c_{57}	−1474611550121835
c_7	4012510	c_{24}	−61151754460109993	c_{41}	625252460652870848640	c_{58}	−752238501145083
c_8	62594937	c_{25}	−21874799137904458	c_{42}	−257915186927068035456	c_{59}	2922618062182023
c_9	−273294726	c_{26}	350144134733804855	c_{43}	−746451546139394837120	c_{60}	−392088201390456
c_{10}	−1295033857	c_{27}	−73723767747911188	c_{44}	524529823070548772352	c_{61}	−305867478309797
c_{11}	9115383412	c_{28}	−1598559526632814545	c_{45}	668277665051029039872	c_{62}	863861891535994
c_{12}	14927382550	c_{29}	1085486427466033552	c_{46}	−703624600512502736896	c_{63}	114026284119490
c_{13}	−199328772012	c_{30}	5826270653461905305	c_{47}	−412842135940581740544	c_{64}	−64040227978608
c_{14}	−12939514508	c_{31}	−6523995000424148584	c_{48}	679347827710005467136	c_{65}	26328579021209
c_{15}	3124239633448	c_{32}	−16790701837856726926	c_{49}	129142322776994562048	c_{66}	14392435408896
c_{16}	−3081873000815	c_{33}	26994175769053333498	c_{50}	−478146168528885407744	c_{67}	−1546188226560
c_{17}	−36462886510382	c_{34}	37210719759516741396	c_{51}	39950117355248222208		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 67, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $67 + 4$ terms removed returns order 67 as well, so $\deg q_{\min} = 67 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $67 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 67 that this sequence satisfies — and that family turns out to have exactly one member.

References

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